

VOLUME VII · GO / NO-GO · VERIFIED AGAINST THE REPOSITORY

Soil Lab Readiness Check

Is it all implemented, and could a real soil laboratory run on it tomorrow?
No, and not yet. The security and data-integrity work is genuinely done.
The catalogue — the substance of Volumes III and VI — has not been built,
and three things would go wrong on the first day.

CHECKED	PRODUCTION	IMPLEMENTED
soilfer-lims @ 416c201 · 2 Sep 2026	Live · build deployed today 08:36 UTC	Volumes I & II · Volume III not started
TO PILOT-READY		
~3 weeks · to seven labs ~10		

What is done is done properly. Every critical finding from Volume I is closed and verified — deny-by-default scoping, the user-management escalation paths, SIS key scoping, deactivation actually deactivating. Volume II's Phases A to D are real: the Result table is append-only with method, replicate, censoring, basis and QC-batch fields; the certificate no longer claims an oven-dry basis it cannot support; interpretation normalises by unit and refuses unknown ones; the decimal-comma defect is gone; texture class derives correctly. That is substantial and it is the hard half.

The catalogue has not been built. The commit titled "seed exhaustive 10-domain analytical catalogue with 166 parameters, 88 methodologies and 23 standard packages" changes exactly two files — analysisService.js and analysisNames.js — and what it adds is FALLBACK_ANALYSIS_NAMES, a code-to-display-name lookup. It is a genuine and useful UI improvement. It is not a catalogue: no units, no methodologies, no standard references, no detection limits, no plausible ranges, no per-lab defaults, and nothing in the database.

There is still no Unit table, no MethodReference table and no LabMethodDefault.

Everything I flagged as outstanding two checks ago is still outstanding.

tests/setup.js is still absent, so eleven test suites still cannot run.

validateSampleMatrix still has zero call sites. Puppeteer is still not installed.

Equipment routes are still ungated. The login limiter is still 100 per fifteen minutes. Nothing has regressed — but nothing on that list has moved either.

36 CLOSED Volumes I & II — verified in source	3 DAY-ONE BLOCKERS Wrong output or lost data on first use	0 / 214 CATALOGUE Not seeded; names only	11 SUITES STILL DEAD No test can run, no CI only
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01 Could a laboratory actually work on it?

Assessed by what a soil laboratory does between a sample arriving and a certificate leaving. This is the question that matters more than any finding count.

LABORATORY FUNCTION	STATUS	WHAT HOLDS OR FAILS
Receive & register samples	READY	Intake workflow, operational gates, barcodes, KoboToolbox field sync with the depth fix. This part is solid.
Assign work	READY	Work items, technician queues, batches, equipment logging. RBAC and lab isolation verified fixed.
Enter results	PARTIAL	The record is excellent — append-only, method-bound, censoring, basis, replicates. But every analysis must be hand-created with a free-text unit first, and no cross-parameter check runs.
Quality control	NOT READY	Batch.qcResults is still an unstructured JSON string. No certified values, no recovery, no RPD, no pass/fail, no proficiency testing. A failed batch cannot invalidate its results.

Review & approve	READY	Manager queue, submissions, re-analysis requests, review decisions, and now a real audit trail behind them.
Issue the certificate	PARTIAL	Structure and basis statement are right. PDF download returns JSON with HTTP 200. Interpretation fires only for six hard-coded codes and is silent for everything else.
Traceability & audit	READY	Append-only results, supersession, audit log, impersonation with actor claim. This would survive scrutiny.
Multi-laboratory isolation	READY	Deny-by-default scoping, lab-scoped WebSockets, scoped SIS keys. Verified in source.
Configure the laboratory	NOT READY	No catalogue, no unit vocabulary, no SOP library, no per-lab method defaults, no setup wizard. Configuration is a week of typing per laboratory, and errors are silent.
Change safely	NOT READY	No tests can run and there is no CI. Any of the above can regress without anyone noticing.



02

The three day-one blockers

BLK-1

BLOCKER

Without the catalogue, the safety work you already paid for does nothing

This is the one that surprised me. Volume II's Phase A made interpretation unit-aware and, crucially, made it *refuse* to interpret an unrecognised unit — which was the correct fix. But a laboratory still types its units by hand into a free-text field. Type g/Kg, gkg, G/KG or ppm and the engine correctly refuses, so the certificate silently loses its interpretation column and nobody is told why.

The same applies to the six hard-coded threshold keys. Unless the laboratory happens to code phosphorus exactly AVAIL_P and pH exactly PH_H2O, no interpretation appears at all. A

laboratory would reasonably conclude the feature is broken.

FIX

Seed the catalogue with a controlled unit vocabulary. This single item converts several already-completed fixes from dormant to working, which is why it is first rather than fifth.

BLK-2 BLOCKER

Quality control cannot be recorded, so the laboratory cannot demonstrate anything

A soil laboratory runs blanks, duplicates and a control sample with every batch — not as good practice but as the basis on which its numbers mean anything. Today those go into `Batch.qcResults` as free text. Nothing computes whether a control passed, nothing charts it over time, and because `Result` has a `batchId` but no disposition logic, a failed batch leaves its results in circulation, indistinguishable from good ones.

For SoilFER this is also a reportable indicator — Activity 1.1's means of verification explicitly lists proficiency testing results, and there is no model to hold one.

FIX

Typed QC records with certified value, window and computed outcome, plus a disposition that flags the batch's results. The minimum viable version is three record types and one pass/fail rule.

BLK-3 BLOCKER

The public report PDF still returns JSON, with a 200

Flagged in Volume I, unchanged since. `puppeteer` appears zero times in `package.json`, so `require('puppeteer')` always throws, the catch always fires, and the endpoint returns `{ format: 'pdf-data', ... }` with HTTP 200 and a console warning nobody reads. The client has no handler for that shape.

A client who receives a shared report link and clicks "Download PDF" gets a broken file and no error. It is the most visible defect a laboratory's own customers would meet, and the fix is a dependency and a Dockerfile line.

FIX

Add `puppeteer-core` plus Alpine `chromium`; make the missing-browser branch a 503 so it can never fail silently again.

03

What is genuinely finished

Worth stating plainly, because the list above is long and the work behind this one was harder.

- **All nine Volume I criticals, verified.** `buildScopedWhere` denies rather than returning `{}`; the spectral library is scoped; `getUsers` combines with AND and honours `?labId=` only for super admins; `updateUser` uses a strict allowlist; SIS keys intersect rather than being overridden; `isActive` is checked at login, on every request and at the WebSocket handshake.
- **The result record is now defensible.** Append-only with `methodologyId`, `replicateNo`, `isCurrent`, `supersededBy`, `censoring`, `basis`, `enteredBy`, `analysedAt`, `equipmentId` and `batchId`. No upsert remains. Re-analysis supersedes rather than destroys.
- **The certificate no longer says anything untrue.** The oven-dry claim is now an air-dry fine-earth statement that matches what is actually measured.
- **Lab-scoped realtime.** Zero `broadcastToAll` calls remain in the controllers; `broadcastToLab` replaced them. The cross-lab refetch storms are gone.
- **KoboToolbox is properly integrated.** A real scheduler, timeouts, pagination, exponential-backoff retry, and depth-aware identifier matching that no longer collides topsoil with subsoil.
- **Human-readable names throughout.** The most recent commits map 166 analysis codes to proper display names and render them across the UI. That is exactly the right instinct — it is the catalogue's presentation layer arriving before the catalogue.

04

Still open, by volume

ITEM	STATUS	EVIDENCE
Catalogue seeded Vol III + VI · 214 analyses	NOT STARTED	No Unit, MethodReference or LabMethodDefault models; names-only lookup in <code>analysisService.js</code>
Sample matrix fields	NOT STARTED	No matrix, module, plantPart or growthStage in

Vol VI

the schema

Setup wizard & per-lab defaults Vol III-2	NOT STARTED	—
Typed QC Vol II QC-01	NOT STARTED	No certifiedValue, recovery or rpd anywhere in the schema
Proficiency testing Vol IV OBL-01 · indicator	NOT STARTED	No model
Cross-parameter checks wired Vol II	NOT WIRED	validateSampleMatrix — 0 call sites
Test harness Vol I QUAL-01	MISSING	server/tests/setup.js absent; 11 suites cannot import
CI Vol I OPS-02	MISSING	Only deploy-docs.yml
PDF generation Vol I INT-03	BROKEN	puppeteer — 0 occurrences in package.json
Token revocation Vol I AUTH-04	MISSING	tokenVersion — 0 occurrences
Equipment authorization Vol I RBAC-06	MISSING	0 checkPermission in equipmentRoutes.js
Login rate limit Vol I AUTH-05	UNCHANGED	Still max: 100 per 15 min across all of /api/auth
MIR spectral window Vol II SPC-02	UNCHANGED	minWavenumber: 600
Result provenance Vol IV ADD-04	MISSING	No provenance field on Result
Profile separation Vol V	NOT STARTED	No profiles/; LAB_CONFIG and LAB_MAPPING still hard-coded; workflow DAG still in JavaScript
GloSIS mapping shelf Vol IV COR-01	NOT STARTED	Ten dormant glosis_* columns still on Analysis and Methodology

05 Shortest path to one real laboratory

Not the full roadmap — the minimum that lets a single friendly laboratory run real samples safely. Roughly three weeks. Everything else can follow while they are using it.

- 2 days **Write tests/setup.js and turn on CI.** Eleven suites start running, including the RBAC suite. Do this first so the next three weeks cannot silently break what Volumes I and II fixed.

- 1 day **Fix the PDF.** puppeteer-core plus chromium in the image; 503 instead of a silent 200.

- 1 day **Close the small security items.** Equipment route guard, tokenVersion, login limiter. An hour each, and they stop being on every future list.

- 5 days **Seed the unit vocabulary, the reference library, and the analyses that laboratory actually runs.** Not all 214 — the twenty-five to forty on their price list, with units, methods, references, decimals and ranges. This is what makes the interpretation and precision work already built start functioning.

- 1 day **Wire validateSampleMatrix.** Written, tested by nothing, called by nothing. Texture closure and C:N plausibility start working the moment it is invoked on submission.

- 4 days **Minimum viable QC.** Three typed record types — blank, duplicate, control — with an expected value, a window and a computed pass or fail, plus a batch disposition that flags its results. Proficiency testing can wait; this cannot.

- 2 days **Per-lab method defaults and a short setup screen.** Enough that a technician never picks a method and a manager can change one. The twelve-question wizard can come later.

- 2 days **Run a full mock batch end to end.** Twenty real samples, intake to signed certificate, with a QC batch and one deliberate re-analysis. This finds what no audit will.

After that: the remaining catalogue and matrices, proficiency testing, the profile separation, and the NSIS interface — in the Volume V order. Pilot in about three weeks, seven laboratories in about ten, generic release after a full sampling campaign.

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06 One thing worth saying directly

The pattern to watch

- **Three times now, correct code has been written and never called.** `checkScope` in Volume I. `validateSampleMatrix` in Volume II, still at zero call sites. And now a 166-entry analysis map that looks like a catalogue in the commit message and is a display-name lookup in the source. Each is good work. None of it is reachable.